

ABSTRAK

Penelitian ini bertujuan untuk: (1) mengembangkan media pembelajaran *Industrial Internet of Things* (IIoT) berbasis *Virtual Smart Factory* pada mata kuliah Praktik Pemrograman Aplikasi Industri / SCADA di Program Studi Pendidikan Teknik Mekatronika, (2) mengetahui tingkat kelayakan media pembelajaran IIoT berbasis *Virtual Smart Factory* berdasarkan penilaian ahli media dan ahli materi, (3) mengetahui respons mahasiswa sebagai pengguna terhadap media pembelajaran IIoT berbasis *Virtual Smart Factory* yang dikembangkan.

Penelitian ini merupakan penelitian pengembangan (R&D) menggunakan model ADDIE (*Analysis, Design, Development, Implementation, Evaluation*), dilaksanakan pada bulan Maret–Juni 2026 di Program Studi Pendidikan Teknik Mekatronika, Fakultas Teknik, Universitas Negeri Yogyakarta. Sumber data terdiri atas ahli materi, ahli media, dan 32 mahasiswa. Teknik pengumpulan data menggunakan angket validasi ahli dan angket respons pengguna. Validitas instrumen diuji melalui *construct validity (expert judgment)*, sedangkan reliabilitas diuji dengan koefisien *Cronbach's Alpha* sebesar 0,878 berkategori sangat tinggi. Data dianalisis menggunakan teknik deskriptif kuantitatif berupa perhitungan rerata skor kelayakan.

Hasil penelitian menunjukkan media berhasil dikembangkan dalam bentuk ekosistem simulasi *digital twin* yang mengintegrasikan *SoftPLC* TIA Portal, Factory I/O, *gateway* Node-RED, dan basis data InfluxDB, dilengkapi Buku Modul Panduan dan 5 seri *labsheet* praktikum. Validasi ahli media memperoleh rerata skor 65 dan ahli materi 80, keduanya berkategori Sangat Layak, sedangkan respons 32 mahasiswa memperoleh rerata skor 98,2 berkategori Sangat Baik. Media disimpulkan layak dan praktis untuk memfasilitasi praktik arsitektur IIoT *end-to-end* tanpa perangkat keras fisik. Media disarankan dikembangkan dengan variasi skenario produksi dan integrasi IIoT berbasis *cloud*.

Kata Kunci: *Digital Twin, Industrial Internet of Things (IIoT), Media Pembelajaran, Simulasi, Virtual Smart Factory.*

**DEVELOPMENT OF IIOT LEARNING MEDIA BASED ON VIRTUAL
SMART FACTORY IN THE INDUSTRIAL APPLICATION PROGRAMMING
/ SCADA PRACTICUM COURSE**

**Farhan Yuda Mahendra
NIM. 22518244007**

ABSTRACT

This study aims to: (1) develop an Industrial Internet of Things (IIoT) learning media based on a Virtual Smart Factory for the Industrial Application Programming Practice / SCADA course in the Mechatronics Engineering Education Study Program, (2) determine the feasibility level of the media based on expert assessments, (3) determine student responses to the developed media.

This study is a research and development (R&D) study using the ADDIE model (Analysis, Design, Development, Implementation, Evaluation), conducted from March to June 2026 at the Mechatronics Engineering Education Study Program, Faculty of Engineering, Universitas Negeri Yogyakarta. Data sources consisted of material experts, media experts, and 32 students, collected through expert validation and user response questionnaires. Instrument validity was tested through construct validity (expert judgment), while reliability was tested using Cronbach's Alpha coefficient of 0.878, categorized as very high. Data were analyzed descriptively using mean feasibility scores.

The results showed that the media was successfully developed as a digital twin simulation ecosystem integrating SoftPLC TIA Portal, Factory I/O, a Node-RED gateway, and an InfluxDB database, equipped with a Guidebook and 5 practicum worksheets. Expert validation scored 65 for media and 80 for material, both categorized as Very Feasible, while 32 students responded with a mean score of 98.2, categorized as Very Good. The media is concluded to be feasible and practical for facilitating end-to-end IIoT practice without physical hardware, and is recommended for further development with varied production scenarios and cloud-based IIoT integration.

Keywords: *Digital Twin, Industrial Internet of Things (IIoT), Learning Media, Simulation, Virtual Smart Factory.*